

Choose Rock for the Job

BUILD • Science • 40 minutes

I can choose a supplied rock sample for a stated model job and justify the decision with relevant property evidence and a limit.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
criteria	the job rules
candidate	one possible choice
justify	give evidence for why

First idea

For a rain-cover model, Sample R lets 0.5 mL through and forms flat pieces. Sample P lets 1.0 mL through but has the faintest scratch mark. Which is the stronger first choice for this brief?

- A. Sample R, because its lower water passage and flat overlapping shape answer this brief.
- B. Sample P, because scratch resistance must decide every job.
- C. Choose whichever colour the group likes most.

My first idea and reason:

Rain-cover for the learning yard

Needs: low water passage • some scratch resistance

appearance does not matter here

The job brief

We do • choose, explain, check

For a hard-wearing path model, which decision uses the most relevant supplied evidence?

A. Choose P because it had a faint scratch mark and the surface card shows a stable, even path piece; keep grip as a separate safety check.

B. Choose D because its pattern is the most colourful.

C. Choose R because it let the least water through, even though the brief is about wear and walking surface.

My choice: _____ Evidence or reason:

Rock-choice evidence matrix

Route each candidate evidence card to RAIN COVER, PATH, INDOOR DISPLAY or NEEDS ANOTHER CHECK.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
RAIN-COVER MODEL	
HARD-WEARING PATH MODEL	
INDOOR DISPLAY	
NEEDS ANOTHER CHECK	

Activity cards

1 • Candidate R	2 • R caution
0.5 mL through in the fixed water test; source card says it can form flat overlapping pieces; moderate scratch mark.	The 120-second result does not prove permanent waterproofing or real roof approval.

3 • Candidate P	4 • P safety gap
Faintest mark in the fixed scratch comparison; supplied surface card shows an even rough path piece; 1.0 mL through.	No classroom result approves grip, loading or installation; those remain separate checks.
5 • Candidate D	6 • Colour-only vote
Supplied piece has a clear pattern and smooth finish; it is stable indoors but shows a deeper scratch mark and 3.5 mL water passage.	Choose D for the outdoor path because most pupils like its colour.
7 • Candidate U	8 • Display stability
U looks smooth in a photograph, but no scratch, water, shape or source evidence is supplied.	D may fit the visual brief only if the supplied base remains stable and is used indoors as stated.

Say: I chose ___ for the model ___ because ___. Candidate ___ was weaker because ___. The matrix cannot tell us ___.

Your lesson record

Choose a supplied candidate for one model job, compare at least two options and revise or confirm the decision after a genuine audience question.

Model helps us see

Use: The brief needs ___. I compared ___ and ___. I chose ___ because ___. We still need to check ___.

Model does not show

The fixed matrix is fictional and simplified. It supports classroom reasoning only; it cannot approve real materials, structures, costs, installation or safety.

Supported

Choose between two candidates for one brief and complete an oral, pointed or scribed because sentence plus one question-mark limit.

Use: The job needs ___. I choose ___ because ___. We still need to check ___.

Standard

Compare three candidates on relevant criteria, justify one choice and revise or confirm it after an audience question.

Cross out irrelevant columns before comparing.

Stretch

Evaluate a trade-off, defend why one tempting candidate is unsuitable and explain how the decision would change for a different purpose.

Keep classroom evidence separate from real construction approval.

Evidence target

Brief criteria, two-candidate comparison, chosen candidate, relevant because evidence, one limit, one audience question and visible response.

Exit, Audience and evidence • W12B

Supported: Complete or point: For the model ___, I chose ___ because ___.

Standard: State your choice, strongest evidence and one limit after the audience response.

Stretch: Explain whether the audience question changed, confirmed or transferred your decision and why.

What I said, and what it changed

Next I will _____.

Adult witness note

What was observed, and with what level of independence?

My evidence and explanation

What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.

Code	Scratch	Water (120 s)	Look
A	line gone	0.0 mL	speckled
C	groove	8.0 mL	sandy
D	faint groove	1.5 mL	pale
F	stopped	9.5 mL	holey

The evidence matrix for four codes

Code	Scratch	Water (120 s)	Look
A	line gone	0.0 mL	speckled
C	groove	8.0 mL	sandy
D	faint groove	1.5 mL	pale
F	stopped	9.5 mL	holey

Why not D?" — a fair question. Answer with a column

Code A highlighted with an audience question

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Evidence matrix

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
